

CS F407: Search and A*

AI Laboratory

Task 0: Problem formulation

Component	Specification
S	Traversable grid positions (r, c)
A	Up, down, left, right
T	Move one cell if it is in bounds and traversable
s_0	$(1, 1)$
G	$\{(7, 15)\}$
c	One per legal move

Only the robot's position changes, so the map need not be part of each state. Walls and out-of-bounds cells make actions invalid. Transitions are deterministic. A solution is a finite legal path from S to G .

Tasks 1–4: Agent, A*, and tests

The grid is a tuple of strings; positions are integer pairs. `neighbours` implements actions and transitions. `astar` tests the goal, stores best known g values in `distance`, computes h with `heuristic`, and places $f = g + h$ in a heap. Heap ties are broken by insertion order. `parent` reconstructs the path. Improved costs replace previous entries; stale heap entries are skipped. States can be reopened. Movement order is up, down, left, right. Expanded counts successor generation events, excluding the goal and including re-expansions.

Test	Found	A* moves	A* expanded
original	Yes	40	63
adjacent	Yes	1	1
blocked	No	–	9
alternative paths	Yes	6	11

All paths were replayed against the grid. BFS independently checked shortest lengths for the admissible heuristics. **Original A* route.** The grid overlay is in `path.txt`; full coordinates are in `results.json`.

Tasks 5–6: Search and heuristic comparison

Original-map algorithm	Moves	Expanded
BFS	40	63
A*: Manhattan	40	63
$h = 0$	40	63
A*: Euclidean	40	63
$2 \times$ Manhattan	40	66

BFS and Manhattan A* both return 40 moves and expand 63 states. The heuristic does not reduce exploration here. Manhattan distance is a lower bound because each step changes one coordinate by one, and obstacles cannot shorten the displacement. Euclidean distance is also a lower bound but usually gives less guidance. With $h = 0$, A* becomes uniform-cost search, equivalent to BFS for unit costs under the same insertion order. Twice Manhattan is not generally admissible. It returns 40 moves here but expands 66 events because improved routes reopen states. This observation does not imply shortest paths on other maps. An aggressive heuristic can return an expensive route before exploring a cheaper one.

Task 7 and final reflections

The first executed implementation passed all four tests. The heap, best-cost dictionary, and predecessors correspond to the design. No repair was required after these tests. Known-answer cases and the BFS comparison were more useful than visual plausibility. Problem formulation specifies legal behaviour before implementation. A* is informed because it estimates remaining cost in addition to cost incurred. Heuristic quality changes exploration and optimality guarantees. The LLM supplied the implementation; executable checks verified legality, reachability, and shortest lengths on these cases. Without testing, stale entries, invalid moves, or incorrect path reconstruction could go undetected.

Reproduce. Run `../venv/bin/python solution.py` here. Results: `results.json`; prompt and correction record: `prompts.txt`.